

Sample End Semester Question Paper - 2025

Subject: Machine Learning | Code: AIML401 | Semester: 4

Branch: AI & ML | Regulation: R20 (sample) | Time: 3 Hours | Max Mar

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. Define a hypothesis.
2. What is a loss function?
3. Name one clustering algorithm.
4. What is a decision boundary?
5. State one use of cross-validation.
6. Define machine learning.
7. Name two supervised learning algorithms.
8. What is a classifier?
9. State one evaluation metric.
10. What is a feature?

PART B - Long questions (6 x 13 = 78 Marks)

1. Explain the ML paradigm: supervised, unsupervised and reinforcement with examples.
2. Explain linear and logistic regression with derivation.
3. Explain k-NN with a worked example.
4. Explain decision trees: information gain and construction.
5. Explain train/test split, cross-validation and evaluation metrics.
6. Explain overfitting, underfitting and regularisation.